Technology Stack

Date	1 July 2026
Team ID	xxxxxx
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*HTML5, CSS3, JavaScript*	*HTML5 provides the structure of the web pages, CSS3 creates a responsive and visually appealing user interface, and JavaScript improves user interaction through client-side validation and dynamic behavior . Together, they provide a simple and user-friendly interface for entering agricultural parameters and displaying crop prediction results.*

2	**Backend / Server-Side**	*Python, Flask*	*Python is widely used for Machine Learning applications because of its extensive libraries and ease of development. Flask is a lightweight web framework that efficiently connects the frontend with the trained Machine Learning model, processes user requests, and returns crop prediction results quickly.*
3	**Database / Data Storage**	*CSV Dataset,* *Pickle (.pkl)* *Model*	The agricultural dataset is stored in CSV format for easy preprocessing and analysis. The trained Machine Learning model is saved as a Pickle (.pkl) file, enabling faster loading and real-time crop predictions without retraining the model.
4	**Cloud / Hosting / Deployment**	*Local Flask Server (Future: Render/Heroku)*	The agricultural dataset is stored in CSV format for easy preprocessing and analysis. The trained Machine Learning model is saved as a Pickle (.pkl) file, enabling faster loading and real-time crop predictions without retraining the model.
5	**Version Control & CI/CD**	Git, GitHub	*Git is used to track source code changes and manage different versions of the project. GitHub provides a centralized repository for collaboration, backup, documentation, and project sharedenabling efficient team development and future enhancements.*

6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy , Matplotlib	Scikit-learn is used for training and implementing Machine Learning algorithms. Pandas and NumPy perform efficient data cleaning, preprocessing, and numerical computations. Matplotlib is used to visualize agricultural data and model performance during the development and analysis stages of the project.